

Midterm Blueprint - Multi-Agent Workforce Intelligence System for
the Energy Industry

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ITAI 2376 Deep Learning Artificial Intelligence

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April 6th, 2026

1. Problem Statement

I work as a People Strategy Lead for ABB's Energy Industries division in Houston. On a daily basis, it includes workforce planning, HR business partnering, and overseeing staffing in seven countries across the Americas. A few weeks ago, I attended CERAWeek 2026: four days, 10,000 energy leaders, 14 sessions. The strongest message was not about gigawatts or gas turbines but about people. The energy industry doesn't have enough workers to build and operate what it's planning, and honestly nobody in the room had a credible plan to fix it.

The IEA's World Energy Employment 2025 report shows 76 million energy jobs worldwide, but in advanced economies, there are 2.4 workers nearing retirement for every new hire under 25. At CERAWeek, several panels cited that 40% of plant operators could retire by the end of this decade, and 42% of people who have left energy since 2016 didn't just switch companies. They left the sector entirely for manufacturing or other industries like software. One company reported 8,000 open craft positions. And now, data centers are a new competitor for those same workers. McKinsey estimates that \$7 trillion will flow into data center construction over five years. IEEE Spectrum reported that the data center boom is creating massive demand for skilled engineers and technicians. The Associated Builders and Contractors estimates that 349,000 new construction workers will be needed in 2026 alone to meet demand. These projects need electricians, welders, and power systems engineers, the same people we actually require.

On the AI front, Accenture announced at CERAWeek that 85% of companies believe AI will transform their business, but only 18% are actually seeing benefits. In energy, only 16% qualify as "talent reinventors." Just 8% of energy organizations know their workforce's skills. We're losing experienced workers to retirement, younger employees to other industries, and craft workers to data centers, while lacking visibility into remaining capabilities. That's the core issue. I'm developing two AI tools: one analyzes resumes to match candidates with specialized energy roles through natural language understanding rather than simple keyword searches, and the other forecasts potential attrition and hiring gaps over the next 6–12 months. The output is a dashboard that shows an HR team both who to hire and where the gaps are forming before they become emergencies.

2. Option Choice and Justification

I'm choosing Option B = Multi-Agent System. Matching a candidate to a job posting and predicting workforce attrition are two very different problems. One is a language task: you have unstructured text (such as a resume or job description) and need a model that can interpret it to determine if the person's experience aligns with the role's requirements. The other is time-series, you have months of headcount data, turnover records, economic indicators, and you need to spot the patterns that lead to attrition.

I figured that combining both into a single agent would make it more difficult to debug. With two agents, I can retrain the matching model with new energy terminology without impacting the forecasting pipeline. It also illustrates how HR teams truly function, with recruiters and workforce planners as distinct roles that exchange data. CrewAI supports exactly that kind of role-based setup.

3. Agent Architecture

The diagram below illustrates the complete system following the INPUT → DL MODEL → AGENT BRAIN (LLM) → TOOL → OUTPUT flow for each agent. Since this is Option B, each agent has its own full pipeline, connected through a CrewAI orchestrator in the center. Every arrow shows what is passed between the boxes.

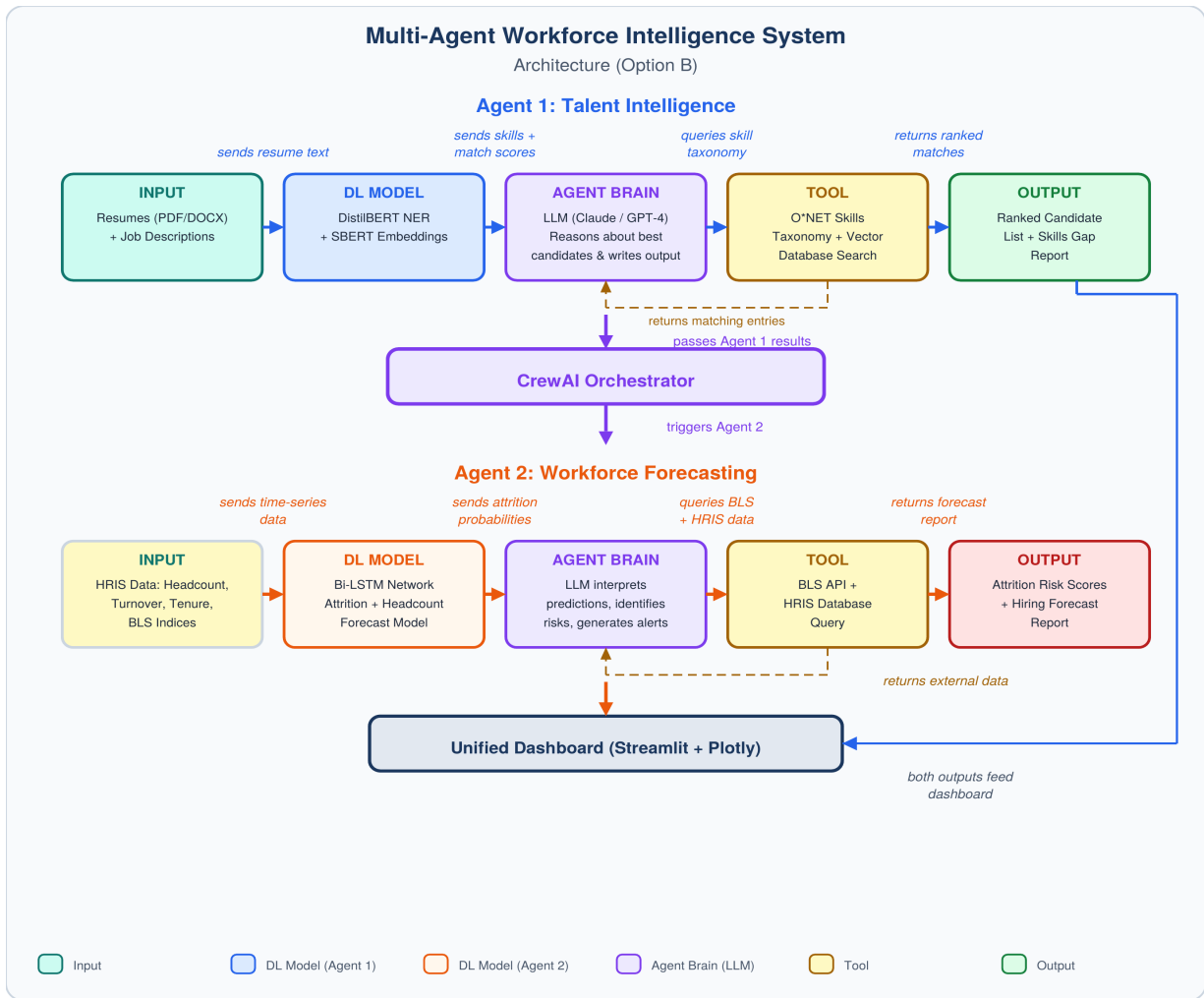


Figure 1: System architecture with inputs, agents, orchestration, and outputs.

3.1 Agent 1 - Talent Intelligence Agent. Ingests resumes (PDF/DOCX) and job descriptions. Runs a fine-tuned DistilBERT model for Named Entity Recognition to extract skills, certifications, years of experience, and past employers from unstructured resume text. Then encodes both the candidate profile and the job requirements into vector embeddings using Sentence-BERT (SBERT), calculating cosine similarity to determine the match. The LLM brain (Claude or GPT) receives the NER-extracted entities and SBERT match scores, reasons about which candidates best fit the role requirements, identifies specific skills gaps for each candidate, queries the O*NET taxonomy for standardized skill mappings, and generates the final ranked report with actionable hiring recommendations. The output is a ranked list of candidates with a skills-gap breakdown. This directly addresses the “8% skills visibility” problem that Accenture highlighted at

CERAWeek. If you don't know what skills your people have, you can't plan anything.

3.2 Agent 2 - Workforce Forecasting Agent. Takes in historical HRIS data, including monthly headcount, termination records, tenure distributions, comp ratios, performance ratings, and external signals such as BLS energy-sector employment indices. Feeds all of it into a Bi-LSTM network that predicts individual attrition probability (binary: will this person leave within 6 months) and department-level headcount forecasts (regression: where will we be at 3, 6, and 12 months out). The LLM brain receives these raw predictions, interprets which departments face the highest risk, queries the BLS API and HRIS database for additional context, identifies whether attrition is driven by compensation gaps, retirement cohorts, or market competition, and generates prioritized alerts with specific hiring recommendations. This is the tool that could have told Sempra 3.5 years before their LNG project opened that they needed to start building the operations team, instead of relying on one leader's foresight.

3.3 CrewAI Orchestrator. Coordinates the two agents. When a new req opens, it triggers Agent 1 to score available candidates and simultaneously asks Agent 2 for the attrition forecast for that role family. Both results feed a unified dashboard. The idea came from something I kept hearing at CERAWeek. The companies getting value from AI are the ones that integrated it into their workflows instead of bolting it on top of old processes. That's the 85/18 gap = 85% believe, 18% benefit.

4. Deep Learning Connection

Transformers (Agent 1)

The Transformer from the 2017 paper "Attention Is All You Need" is what Agent 1 runs on. The key mechanism we covered in class is self-attention. Instead of processing a sentence from left to right, a Transformer looks at every word in relation to every other word at the same time. The attention weights determine which parts of a sequence are relevant to which other parts. This is especially important for resume parsing. For example, a term like "Python." On a controls engineer's resume, "Python" next to "SCADA integration" means something entirely different than "Python" next to "web scraping." Our ATS treats them equally, but a Transformer does not because it has learned the contextual

relationships between terms during pre-training. At CERAWeek, Accenture reported that only 8% of energy organizations have real skills visibility, and I believe this is partly why. The systems we use to catalog skills still match based on keywords, not actual meaning. I'm using DistilBERT for the NER task. In Lab 05, I fine-tuned DistilBERT on AG News, which achieved 90.15% accuracy in 3 epochs, much higher than LSTM at 79.45% and GRU at 81.15% on the same data. That's the advantage of pre-training; after just one epoch, DistilBERT was already at 89.50%, outperforming GRU after five epochs. According to Sanh et al., DistilBERT retains about 97% of BERT's understanding while running 60% faster and using 40% less memory, which matters when processing hundreds of resumes. I plan to fine-tune it on a labeled resume dataset with entity types like SKILL, CERTIFICATION, DEGREE, EMPLOYER, and YEARS_EXPERIENCE. For matching, Sentence-BERT creates fixed-length embeddings for both the candidate profile and the job posting, and I calculate cosine similarity. A 2025 IEEE paper tested this on 900 resumes across various professions, and it performed well. Another study combined SBERT with DeBERTa and got strong matching scores.

RNNs / LSTM (Agent 2)

The vanishing gradient problem is something I could actually see in my Lab 05 results. My Vanilla RNN barely hit 70.30% on AG News because it couldn't hold onto information from more than a few steps back. When you backpropagate through a long sequence, the gradients just wash out. LSTM reached 79.45% and GRU scored 81.15% on the same data, a 10-point jump that shows what happens when you add memory gates. LSTMs use three gates (input, forget, output) to control what the cell state stores, discards, or passes forward. For workforce data, that kind of long-range memory is essential. The signals that predict attrition aren't just in recent numbers. They include seasonal cycles (we often see departures spike in January after bonuses are paid), annual review effects, and long-term trends like retirement cohorts moving through the system over the years. At CERAWeek, one panelist mentioned that 40% of plant operators could retire by the end of the decade. An LSTM trained on the right features should see those cohorts coming and quantify the impact before it hits.

I'm using a Bi-LSTM, which processes the time series in both directions. The forward pass captures how past events lead to attrition, while the backward pass can incorporate knowledge about future events, such as a scheduled plant shutdown. The model accepts multiple inputs, including monthly headcount, termination counts, tenure, pay ratios, engagement scores, and BLS employment indices. It outputs individual attrition probabilities (binary classification) and department-level headcount forecasts (regression). A 2025 LSTNet paper demonstrated that LSTM architectures effectively forecast labor-market trends with multiple economic inputs. MDPI published work confirming deep learning beats random forests and logistic regression for attrition prediction when temporal patterns are present.

5. Agent Framework

I chose CrewAI for orchestration. It lets you define agents with specific roles and goals, assign them tools, and have them collaborate, which maps directly to how I'm structuring the two-agent system. The architecture has two abstractions: Crews (autonomous agent teams) and Flows (event-driven workflows that manage state). I considered LangChain and AutoGen as alternatives. LangChain is a versatile LLM framework, powerful but too heavy for my needs since my agents run dedicated deep learning models, not raw LLM calls. AutoGen is designed for conversational multi-agent setups, which doesn't suit my structured pipeline. CrewAI is lighter, allows custom tool registration so I can wrap PyTorch models as callable tools, and has a large developer community. AWS published production case studies running CrewAI with Bedrock, which gave me confidence it's not just a demo framework. This will be my first time building with CrewAI, but I've reviewed the documentation and run through the quickstart tutorial.

6. Tools & Data

Category	Tool / Source	Purpose
Deep Learning	PyTorch, Hugging Face Transformers	Training & inference: DistilBERT NER, SBERT embeddings, Bi-LSTM forecasting
NLP	spaCy, NLTK	Tokenization, lemmatization, text normalization for resume and job-description input

Agent Framework	CrewAI (open-source)	Multi-agent orchestration, task delegation, inter-agent coordination
Data - Resumes	Kaggle Resume Dataset + synthetic	NER fine-tuning; supplemented with energy-sector resumes for domain terms (P&ID, HAZOP, API 650)
Data - Workforce	IBM HR Analytics (Kaggle), BLS QCEW	Attrition training (IBM); external signals (BLS Quarterly Census of Employment & Wages)
Data - Jobs	O*NET, Rigzone, EnergyJobline	Standardized skill taxonomy and real postings for matching evaluation
Dashboard	Streamlit, Plotly	Interactive UI: match scores, attrition heatmaps, headcount projections
Infrastructure	Python 3.11, Google Colab Pro	Development and GPU training environment

All datasets used in this project are either publicly available (IBM HR Analytics on Kaggle, BLS QCEW, O*NET) or self-generated (synthetic energy-sector resumes created for NER training, and synthesized monthly temporal snapshots derived from the IBM HR cross-sectional data to enable Bi-LSTM time-series training). There are no privacy, licensing, or data-access concerns.

7. Build Plan

Wk	Dates	Deliverables
1	Apr 7 – Apr 13	Repo setup, environment config, dataset preparation (Kaggle resumes, IBM HR attrition, BLS QCEW). Begin hand-labeling ~80–100 energy-sector resumes for NER with domain entities (P&ID, HAZOP, API 650, DCS, SCADA), using spaCy's pre-trained <code>en_core_web_trf</code> as a base and correcting its outputs rather than labeling from scratch. Fine-tune DistilBERT NER on labeled resumes. Agent 1 NER pipeline should extract entities from a sample resume by Friday.
2	Apr 14 – Apr 20	Build SBERT embedding pipeline and cosine-similarity scoring to complete Agent 1 (resume + job description → ranked match score). Simultaneously: synthesize temporal structure from the IBM HR cross-sectional dataset by generating monthly snapshots with realistic turnover distributions, since the original dataset is a single point-in-time snapshot without time-series data. Feature engineering on synthesized temporal data. Build Bi-LSTM in PyTorch, train on attrition classification. Integrate BLS signals as external features. Agent 2 should output individual attrition risk scores by Friday.

3	Apr 21 – Apr 27	Wire both agents into CrewAI Flow. Orchestration logic: trigger both agents on a sample scenario, merge outputs. Build Streamlit dashboard with match table, attrition heatmap, forecast chart. Integration testing.
4	Apr 28 – May 3	Evaluate: F1 for NER, AUC-ROC for attrition, MAE for headcount forecast. End-to-end testing and bug fixes. Write final report. Record demo. Submit all deliverables.

8. Anticipated Challenges

8.1 Not enough labeled energy-sector resumes. This is what worries me most. General NER datasets don't understand what "P&ID review," "HAZOP facilitation," or "API 650 certified" means. These terms distinguish a good candidate from a bad one in energy hiring, yet they're invisible to a model trained on generic data. I plan to use spaCy's pre-trained `en_core_web_trf` as a starting point and manually correct its outputs on 80–100 energy resumes, then use data augmentation methods like synonym swaps and back-translation to expand the dataset. If it's still limited, I'll transfer learn from a pre-trained NER model. The Accenture data from CERAWEEK ("only 8% have skills visibility") tells me this exact problem exists at the industry level, not just in my training data.

8.2 Lopsided attrition data. About 85% of employees stay, and 15% leave in any given time. If you train an LSTM on that raw split, it will just predict "stay" every time, making it appear 85% accurate, which is useless. I faced a similar problem in Lab 05 where the Business category was the hardest to classify because the classes were uneven. I plan to use SMOTE to balance the training set and a weighted loss function that penalizes missed attrition cases more heavily. Evaluation will be AUC-ROC and precision-recall, not accuracy. Additionally, the IBM HR Analytics dataset is a cross-sectional snapshot (1,470 records at a single point in time), not a longitudinal time series. A Bi-LSTM requires sequential monthly data to learn temporal attrition patterns. My plan is to synthesize temporal structure by generating monthly snapshots with realistic turnover distributions derived from the cross-sectional features (tenure, age, job satisfaction trends). This is a known limitation: the forecasting model will demonstrate the architecture and temporal reasoning capability, but production deployment would require actual HRIS time-series data spanning 2–3 years.

8.3 Wiring custom models into CrewAI. CrewAI expects LLM-based agents, but my setup runs PyTorch models. I'll wrap each model as a Python function (input: resume text, output: JSON of extracted skills) and register it as a CrewAI custom tool. The documentation says this is supported. However, I haven't actually built this integration yet, so Week 4 could run into issues. If it does, I'll simplify the orchestration to a sequential pipeline instead of a full CrewAI Flow.

8.4 GPU access and fairness. Both models require GPU training. I'm using Colab Pro, and if I hit session limits, I'll reduce the number of fine-tuning epochs and use smaller LSTM hidden dimensions. In Lab 05, I learned that DistilBERT's optimal learning rate is about 50 times smaller than that of an LSTM because large updates can cause catastrophic forgetting of pre-trained representations. Therefore, I need to be cautious with the learning rate schedule. Regarding fairness, the "asymmetric benefit" finding from CERAWeek, where AI helps weaker performers more than stronger ones applies both ways in a matching system. If my model over-focuses on patterns in training data (such as school names and employer prestige), it could reinforce biases. I will review match-score distributions across demographic groups and report whatever I discover, even if the results are not clean.

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